# Surya Manavalan

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## Experience

**Software Engineer II, Broadridge** — San Francisco, CA, *Mar 2024 - Present*
- Built and manages the backend services for ProxyVote which handles 550 billion shares across 4,500 shareholder meetings annually: proxyvote.com
- Orchestrated a multi-agent harness to rebuild a large legacy admin service from scratch on a modern stack, remediating hundreds of vulnerabilities and retiring years of tech debt.
- Delivered core modules for Broadridge's Workstation Framework, an internal platform for building compliant, enterprise-grade financial applications, powering apps that manage $122M+ in client assets.
- Developed an agentic workflow skill-brain for our cloud agents to provide platform support, triage, and make small bug fixes autonomously across our *React, Node, and Spring Boot* codebases

**Software Engineer, Momentum Optics Startup** — Denver, CO, *Dec 2023 – Mar 2024*
- Built a landing page and custom client dashboard in collaboration with the CEO: momentumoptics.com
- Developed Python cloud functions handling core backend services: Mailchimp newsletter signups, client dashboard authentication, CRM, and Stripe billing.

**Software Engineer Intern, T-Mobile** — Seattle, WA, *May 2022 - Dec 2023*
- Owned observability rehaul across the frontend *and* backend for payment system by implementing OpenTelemetry interfaced with Jaeger and Prometheus.
- Integrated Trustly's banking widget into customer payment flows, allowing direct bank integration to phone plan. Contributing to a projected $20M+/year in credit card fee cost savings.

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## Projects

**MLRagent.com**— Agentic MLR (medical, legal, regulatory) review for pharma professionals, mlragent.com
- Solo built an enterprise AI tool in use by real medical writing professionals*: A* cloud sandboxed claude agent sdk harness and interface for reviewing medical writing deliverables against FDA guidelines
- Multi-agent system for distilling FDA guidelines into an auditable "agent brain": mlragent.com/brain

**entityspace.ai**— Replication and Visual representation of LLM interpretability research, entityspace.ai
- Replicated Anthropic's Assistant Axis paper on Qwen3-1.7B, with an AWS GPU worker extracting activations for each new persona live.
- Built a Three.js app that places personas in 3D by activation projection; 1,300+ visitors, 280+ summons.

**OpenWorldChess.com**— Browser game & automated AI advertising openworldchess.com
- Built and launched Open World Chess, an open-world multiplayer chess game (TypeScript/Node, WebSockets) that has drawn 700+ players, and set up an automated marketing pipeline in which a scheduled agent uses custom skills to script, film, voice and post a new video ad to TikTok every day.
- Automated Tiktok: tiktok.com/@openworldchess

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## Education

**University of Colorado Boulder -** B.S. in Computer Science | Graduated Dec 2023
Relevant Coursework: Full-stack Development, HCI, Software Engineering Methods, Machine Learning, Operating Systems, Computer Systems, Data structures and Algorithms, Databases, Theory of Comp.